

Probability Distribution Proofs

i) Approximating the Poisson Distribution from the Binomial Distribution

The Poisson distribution can be derived as the limiting case of the Binomial distribution.

Conditions for the Approximation:

Let X be a binomial random variable, $X \sim B(n, p)$, representing the number of successes in n independent trials, each with probability of success p . The binomial distribution converges to a Poisson distribution when:

1. The number of trials $n \rightarrow \infty$ (a very large number of trials).
2. The probability of success $p \rightarrow 0$ (success is a rare event).
3. The mean of the distribution remains constant, meaning $np = \lambda$ (or $p = \frac{\lambda}{n}$), where $\lambda > 0$.

Proof:

The probability mass function (PMF) of a Binomial distribution is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$P(X = k) = \frac{n!}{k!(n-k)!} p^k (1 - p)^{n-k}$$

Substitute $p = \frac{\lambda}{n}$ into the equation:

$$P(X = k) = \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

Now, let's expand the factorials and regroup the terms to evaluate the limit as $n \rightarrow \infty$:

$$P(X = k) = \frac{n(n-1)(n-2) \cdots (n-k+1)}{k!} \cdot \frac{\lambda^k}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(1 - \frac{\lambda}{n}\right)^{-k}$$

We can rearrange the first part by pairing each of the k terms in the numerator with one of the k factors of n in the denominator:

$$P(X = k) = \frac{\lambda^k}{k!} \left(\frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \cdots \frac{n-k+1}{n} \right) \left(1 - \frac{\lambda}{n} \right)^n \left(1 - \frac{\lambda}{n} \right)^{-k}$$

Now, we take the limit as $n \rightarrow \infty$:

1. For the grouped fractional terms:

$$\lim_{n \rightarrow \infty} \left(1 \cdot \left(1 - \frac{1}{n} \right) \cdot \left(1 - \frac{2}{n} \right) \cdots \left(1 - \frac{k-1}{n} \right) \right) = 1 \cdot 1 \cdot 1 \cdots 1 = 1$$

(Since k is a fixed finite integer, as $n \rightarrow \infty$, every term like $\frac{c}{n} \rightarrow 0$.)

2. For the term with the exponent n , we use the standard calculus definition of the exponential function $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n = e^x$:

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n} \right)^n = e^{-\lambda}$$

3. For the term with the exponent $-k$:

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n} \right)^{-k} = 1^{-k} = 1$$

Combining these limits, we get:

$$\lim_{n \rightarrow \infty} P(X = k) = \frac{\lambda^k}{k!} \cdot 1 \cdot e^{-\lambda} \cdot 1$$

$$\lim_{n \rightarrow \infty} P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

This resulting formula is exactly the probability mass function of the Poisson distribution.

ii) Approximating the Normal Distribution from the Poisson Distribution

The Normal distribution can be used to approximate the Poisson distribution when the rate parameter λ is sufficiently large. We can prove this using the Moment Generating Function (MGF).

Conditions for the Approximation:

Let $X \sim \text{Poisson}(\lambda)$. The distribution of X approaches a Normal distribution as:

1. The rate parameter $\lambda \rightarrow \infty$.

For a Poisson distribution, the mean is $\mu = \lambda$ and the variance is $\sigma^2 = \lambda$. We define the standardized random variable Z :

$$Z = \frac{X - \mu}{\sigma} = \frac{X - \lambda}{\sqrt{\lambda}}$$

Proof:

The goal is to show that as $\lambda \rightarrow \infty$, the MGF of the standardized variable Z converges to the MGF of the Standard Normal distribution $N(0, 1)$, which is $e^{t^2/2}$.

First, recall the MGF of a Poisson random variable X :

$$M_X(t) = E[e^{tX}] = \exp(\lambda(e^t - 1))$$

Now, let's find the MGF of our standardized variable Z :

$$M_Z(t) = E[e^{tZ}] = E\left[\exp\left(t \frac{X - \lambda}{\sqrt{\lambda}}\right)\right]$$

$$M_Z(t) = E\left[\exp\left(\frac{t}{\sqrt{\lambda}}X - t\sqrt{\lambda}\right)\right]$$

$$M_Z(t) = e^{-t\sqrt{\lambda}} E\left[\exp\left(\frac{t}{\sqrt{\lambda}}X\right)\right]$$

The expectation on the right is simply the MGF of X evaluated at $\frac{t}{\sqrt{\lambda}}$:

$$M_Z(t) = e^{-t\sqrt{\lambda}} M_X\left(\frac{t}{\sqrt{\lambda}}\right)$$

Substitute the known MGF for the Poisson distribution:

$$M_Z(t) = e^{-t\sqrt{\lambda}} \exp \left[\lambda \left(\exp \left(\frac{t}{\sqrt{\lambda}} \right) - 1 \right) \right]$$

$$M_Z(t) = \exp \left[-t\sqrt{\lambda} + \lambda \left(e^{t/\sqrt{\lambda}} - 1 \right) \right]$$

To evaluate this as $\lambda \rightarrow \infty$, we use the Taylor series expansion for the exponential function $e^y = 1 + y + \frac{y^2}{2!} + \frac{y^3}{3!} + \dots$, where $y = \frac{t}{\sqrt{\lambda}}$:

$$e^{t/\sqrt{\lambda}} = 1 + \frac{t}{\sqrt{\lambda}} + \frac{1}{2!} \left(\frac{t}{\sqrt{\lambda}} \right)^2 + \frac{1}{3!} \left(\frac{t}{\sqrt{\lambda}} \right)^3 + \dots$$

$$e^{t/\sqrt{\lambda}} = 1 + \frac{t}{\sqrt{\lambda}} + \frac{t^2}{2\lambda} + O(\lambda^{-3/2})$$

Substitute this expansion back into the exponent of $M_Z(t)$:

$$\text{Exponent} = -t\sqrt{\lambda} + \lambda \left(1 + \frac{t}{\sqrt{\lambda}} + \frac{t^2}{2\lambda} + O(\lambda^{-3/2}) - 1 \right)$$

$$\text{Exponent} = -t\sqrt{\lambda} + \lambda \left(\frac{t}{\sqrt{\lambda}} + \frac{t^2}{2\lambda} + O(\lambda^{-3/2}) \right)$$

$$\text{Exponent} = -t\sqrt{\lambda} + t\sqrt{\lambda} + \frac{t^2}{2} + O(\lambda^{-1/2})$$

The $-t\sqrt{\lambda}$ and $+t\sqrt{\lambda}$ terms cancel out:

$$\text{Exponent} = \frac{t^2}{2} + O(\lambda^{-1/2})$$

So, the MGF of Z is:

$$M_Z(t) = \exp \left(\frac{t^2}{2} + O(\lambda^{-1/2}) \right)$$

Finally, we take the limit as $\lambda \rightarrow \infty$. The term $O(\lambda^{-1/2})$ goes to 0:

$$\lim_{\lambda \rightarrow \infty} M_Z(t) = \exp \left(\frac{t^2}{2} \right)$$

This is exactly the Moment Generating Function of the Standard Normal distribution $N(0, 1)$. By Lèvy's Continuity Theorem, since the MGF of Z

converges to the MGF of the Standard Normal distribution, the distribution of Z converges to the Standard Normal distribution as $\lambda \rightarrow \infty$.

Stirling's Approximation:

For large values of k , Stirling's approximation for factorials states that:

$$k! \approx \sqrt{2\pi k} \left(\frac{k}{e}\right)^k$$

Proof: Poisson to Normal via Stirling's Approximation

Let X be a Poisson random variable with rate parameter λ . Its PMF is:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

We want to observe what happens as $\lambda \rightarrow \infty$. To do this, we standardize the variable k . Let z be the standard score:

$$z = \frac{k - \lambda}{\sqrt{\lambda}}$$

Rearranging this for k , we get:

$$k = \lambda + z\sqrt{\lambda} = \lambda \left(1 + \frac{z}{\sqrt{\lambda}}\right)$$

Step 1: Substitute Stirling's approximation into the PMF.

$$P(X = k) \approx \frac{e^{-\lambda} \lambda^k}{\sqrt{2\pi k} \left(\frac{k}{e}\right)^k}$$

$$P(X = k) \approx \frac{1}{\sqrt{2\pi k}} e^{-\lambda} e^k \left(\frac{\lambda}{k}\right)^k$$

$$P(X = k) \approx \frac{1}{\sqrt{2\pi k}} e^{k-\lambda} \left(\frac{\lambda}{k}\right)^k$$

Step 2: Analyze the coefficient $\frac{1}{\sqrt{2\pi k}}$.

Substitute $k = \lambda + z\sqrt{\lambda}$:

$$\frac{1}{\sqrt{2\pi k}} = \frac{1}{\sqrt{2\pi\lambda \left(1 + \frac{z}{\sqrt{\lambda}}\right)}}$$

As $\lambda \rightarrow \infty$, the term $\frac{z}{\sqrt{\lambda}} \rightarrow 0$. Therefore, this part simplifies to:

$$\frac{1}{\sqrt{2\pi k}} \approx \frac{1}{\sqrt{2\pi\lambda}}$$

Step 3: Analyze the exponential and power terms.

We need to evaluate $e^{k-\lambda} \left(\frac{\lambda}{k}\right)^k$. Let's rewrite $\left(\frac{\lambda}{k}\right)^k$ using base e :

$$\left(\frac{\lambda}{k}\right)^k = \exp\left(k \ln\left(\frac{\lambda}{k}\right)\right) = \exp\left(-k \ln\left(\frac{k}{\lambda}\right)\right)$$

Substitute $k = \lambda + z\sqrt{\lambda}$:

$$\frac{k}{\lambda} = \frac{\lambda + z\sqrt{\lambda}}{\lambda} = 1 + \frac{z}{\sqrt{\lambda}}$$

So, the entire term we need to evaluate becomes:

$$e^{k-\lambda} \left(\frac{\lambda}{k}\right)^k = \exp\left(k - \lambda - k \ln\left(1 + \frac{z}{\sqrt{\lambda}}\right)\right)$$

Substitute $k - \lambda = z\sqrt{\lambda}$ and $k = \lambda + z\sqrt{\lambda}$ into the exponent:

$$\text{Exponent} = z\sqrt{\lambda} - (\lambda + z\sqrt{\lambda}) \ln\left(1 + \frac{z}{\sqrt{\lambda}}\right)$$

Step 4: Taylor expand the natural logarithm.

For small x , the Taylor series expansion is $\ln(1 + x) \approx x - \frac{x^2}{2} + \dots$

Let $x = \frac{z}{\sqrt{\lambda}}$:

$$\ln \left(1 + \frac{z}{\sqrt{\lambda}} \right) \approx \frac{z}{\sqrt{\lambda}} - \frac{z^2}{2\lambda}$$

Substitute this back into the exponent equation:

$$\text{Exponent} \approx z\sqrt{\lambda} - (\lambda + z\sqrt{\lambda}) \left(\frac{z}{\sqrt{\lambda}} - \frac{z^2}{2\lambda} \right)$$

Expand the brackets:

$$\text{Exponent} \approx z\sqrt{\lambda} - \left(\lambda \frac{z}{\sqrt{\lambda}} - \lambda \frac{z^2}{2\lambda} + z\sqrt{\lambda} \frac{z}{\sqrt{\lambda}} - z\sqrt{\lambda} \frac{z^2}{2\lambda} \right)$$

$$\text{Exponent} \approx z\sqrt{\lambda} - \left(z\sqrt{\lambda} - \frac{z^2}{2} + z^2 - \frac{z^3}{2\sqrt{\lambda}} \right)$$

As $\lambda \rightarrow \infty$, the term $\frac{z^3}{2\sqrt{\lambda}} \rightarrow 0$. We can ignore it:

$$\text{Exponent} \approx z\sqrt{\lambda} - z\sqrt{\lambda} + \frac{z^2}{2} - z^2$$

$$\text{Exponent} \approx -\frac{z^2}{2}$$

Step 5: Put it all together.

Combine the simplified coefficient (from Step 2) and the simplified exponent (from Step 4) back into the PMF equation:

$$P(X = k) \approx \frac{1}{\sqrt{2\pi\lambda}} e^{-z^2/2}$$

Step 6: Convert from PMF to PDF.

In a discrete distribution, the distance between consecutive values of k is

$\Delta k = 1$. In our standardized variable $z = \frac{k-\lambda}{\sqrt{\lambda}}$, a step of $\Delta k = 1$

corresponds to a step size of $dz = \frac{1}{\sqrt{\lambda}}$.

We can rewrite our probability as:

$$P(X = k) \approx \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \left(\frac{1}{\sqrt{\lambda}} \right)$$

$$P(z) \approx \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

This shows that the density function $f(z)$ is:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

This is exactly the Probability Density Function of the Standard Normal distribution, $N(0, 1)$.